

Paper Summary:

ReLU-QP: A GPU-Accelerated Quadratic Programming Solver for Model-Predictive Control

(ICRA'24 Paper by Bishop et al.)

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- Model Predictive Control (MPC) requires solving constrained optimization problems at high frequency.
- ReLU-QP proposes a novel GPU-accelerated formulation that reformulates ADMM as a network-like iteration.

Problem Statement

We aim to solve a convex quadratic program of the form:

$$\min_y \quad \frac{1}{2}y^\top Hy + g^\top y \quad \text{subject to} \quad c \leq Gy \leq d$$

where:

- $H \in \mathbb{R}^{n \times n}$ is symmetric positive definite.
- $G \in \mathbb{R}^{m \times n}$ defines m linear inequality constraints.
- $g \in \mathbb{R}^n$, $c, d \in \mathbb{R}^m$.

ADMM Overview

The Alternating Direction Method of Multipliers (ADMM) introduces an auxiliary variable $z = Gy$ to handle constraints.

$$\mathcal{L}_\rho(y, z, \lambda) = \frac{1}{2}y^\top Hy + g^\top y + \lambda^\top (Gy - z) + \frac{\rho}{2}\|Gy - z\|^2$$

ADMM updates:

$$y^{(k+1)} = \left(H + \rho G^\top G\right)^{-1} \left(-g + \rho G^\top z^{(k)} - G^\top \lambda^{(k)}\right)$$

$$z^{(k+1)} = \Pi_{[c,d]}(Gy^{(k+1)} + \rho^{-1}\lambda^{(k)})$$

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho(Gy^{(k+1)} - z^{(k+1)})$$

ReLU-QP: Reformulating ADMM

- ReLU-QP reorders updates: λ is updated before y .
- This leads to an equivalent update that enables a single-step formulation:

$$v^{(k+1)} = \text{clamp}(Wv^{(k)} + b)$$

- Here, $v = [y; z; \lambda]$, and clamp applies box projection to z .

Detailed Matrix Formulation

Let $D = (H + \sigma I + G^\top \rho G)^{-1}$. Then:

$$W = \begin{bmatrix} D(\sigma I - G^\top \rho G) & 2DG^\top \rho & -DG^\top \\ GD(\sigma I - G^\top \rho G) + G & 2GDG^\top \rho - I & -GDG^\top + \rho^{-1} \\ \rho G & -\rho I & I \end{bmatrix},$$
$$b = \begin{bmatrix} -Dg \\ -GDg \\ 0 \end{bmatrix}$$

Choosing σ

- **Scalar** σ is a small regularization constant (e.g., 10^{-3} , 10^{-6}) added to improve the conditioning of the matrix to be inverted.
- σ is chosen empirically and helps ensure numerical stability in the computation of $D = (H + \sigma I + G^\top \rho G)^{-1}$.
- If solving repeated QPs with fixed structure (e.g., in MPC), W and b can be precomputed for fixed σ which never changes over the series of QP solving within an end-to-end task.

Iteration and Convergence

- ReLU-QP applies the same matrix operation repeatedly.
- Still uses residual-based convergence from ADMM:

$$r^{\text{prim}} = Gy - z, \quad r^{\text{dual}} = Hy + g + G^{\top} \lambda$$

- Tolerances $\varepsilon_{\text{prim}}, \varepsilon_{\text{dual}}$ are checked every few iterations.

Precomputable Components

- If H , G , and ρ are fixed, precompute:
 - $D = (H + \sigma I + G^\top \rho G)^{-1}$
 - W , b
 - Clamp bounds for z : $[c, d]$
- Precomputation allows each iteration to be a single matrix-vector multiply and clamp.

Interpretation as Neural Network Layer

- ReLU-QP iteration: $v \leftarrow \text{clamp}(Wv + b)$
- Structurally equivalent to a layer in a feedforward network
- GPU-parallelizable: no branching, pure linear ops and projection

Summary

- ReLU-QP reformulates ADMM to enable GPU acceleration.
- Uses a unified matrix form with clamping.
- Maintains theoretical guarantees of ADMM.
- Demonstrates significant speedups in real-time MPC.

Reference

A. L. Bishop, J. Z. Zhang, S. Gurumurthy, K. Tracy and Z. Manchester,
"ReLU-QP: A GPU-Accelerated Quadratic Programming Solver for Model-Predictive Control,"
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